Department of Computer Science & Engineering and Information Technology

Major Project Proposal (AY 2025-26)

Group No.	15

1. **Project Title :** Explainable Semi-Supervised Learning for Post-Earthquake Building Damage Assessment

2. **Team Members**

S. No.	Roll No.	Name	Mobile No.	Role
1.	221030034	Khushi Bhasin	8307761921	ML Developer & XAI Researcher
2.	221030455	Abhinav Manda	7734985047	SSL Specialist & ML Developer,Team Lead
3.	221030245	Aatesh	7027867614	System Architect & Integration Engineer
4.	221030331	Vanshika Sharma	8595366196	Evaluation Manager

3. **Name of Supervisor (s)**
   Dr. Arvind Kumar
   Assistant Professor (SG)
   Department of CSE & IT

4. **Work Distribution**

S. No.	Roll No.	Work Distribution
1.	221030034	• Dataset Preparation & Literature Review • Implement baseline CNN model using transfer learning • Research XAI techniques • Conduct rigorous testing with labeled/unlabeled datasets.

2.	221030455	• Dataset Preparation & Literature Review • Integrate semi-supervised learning • Integrate chosen methods into the CNN model for interpretability. • Design the system architecture and integrate model + XAI pipeline into a prototype tool
3.	221030245	• Dataset Preparation & Literature Review • Perform hyperparameter tuning and ablation studies. • Generate heatmaps and visual justifications highlighting damage regions. • Manage version control (GitHub) and coordinate integration efforts.
4.	221030331	• Dataset Preparation & Literature Review • Analyze model accuracy, precision, recall, and F1-score. • Evaluate explanation quality • Lead project documentation, prepare final report, and presentation slides.

5. Problem Statement

Post-earthquake building damage assessment plays a crucial role in disaster response, recovery planning, and ensuring public safety. Traditionally, this task is carried out manually by civil engineers and domain experts through on-site visual inspections. Although effective, this process is extremely slow, resource-intensive, and often subjective, as assessments may vary between individuals. During large-scale disasters, where thousands of buildings must be inspected within a short time frame, manual assessment becomes impractical and delays life-saving decisions. Automated solutions powered by computer vision and deep learning have shown potential in accelerating this process by analyzing post-disaster imagery. However, most state-of-the-art computer vision models require large, carefully labeled datasets for training, which are costly and time-consuming to obtain. Expert-labeled earthquake damage datasets are particularly scarce because structural engineers must provide detailed annotations, making large-scale data preparation infeasible for rapid deployment in real-world scenarios.

On the other hand, a vast amount of unlabeled earthquake reconnaissance imagery is available through post-disaster surveys, drone footage, and online repositories. These images remain underutilized by conventional supervised learning methods. Semi-supervised learning (SSL) provides a promising solution by combining a small set of labeled images with a large pool of unlabeled data, thereby improving model generalization and performance. Techniques such as pseudo-labeling can assign provisional labels to unlabeled data, allowing the model to learn more effectively without additional expert intervention. This

approach significantly reduces dependency on costly annotations while still leveraging the richness of available reconnaissance imagery.

Despite the advantages of SSL, one major challenge remains: trust and interpretability. Deep learning models are often considered "black boxes," as they provide predictions without clear reasoning. In critical scenarios such as earthquake damage assessment, stakeholders including engineers, policymakers, and disaster response teams require not just accurate outputs but also transparent explanations of the model's decisions. Without interpretability, decision-makers may hesitate to rely on automated systems, especially in high-stakes environments involving human lives and infrastructure safety. To address this issue, Explainable AI (XAI) methods can be integrated into the workflow. Tools such as Grad-CAM, saliency maps, and feature attribution techniques highlight the specific regions in an image that influenced the model's prediction, thereby providing valuable visual justifications. For instance, if a building is classified as "severely damaged," the explanation can point to visible cracks, collapsed structures, or tilted columns as evidence, making the system's decision more transparent and reliable.

6. Main Objectives

1) To develop a **semi-supervised learning pipeline**

2) To integrate **XAI methods**

3) To design and implement a **prototype tool** that can process earthquake imagery, output predictions, and highlight critical damaged regions for end-users.

7. Resources Required

Category	Description	
Software Resources	• Python (TensorFlow/Keras, PyTorch) • OpenCV for image preprocessing • Libraries for XAI • Jupyter Notebook / Colab	Version: 3.12.5 Version: 4.10.0 Version: 1.5.0 Version: 7.2.1
Hardware Resources	• GPU-enabled machine	

Others	• Additional unlabeled earthquake reconnaissance images • xBD Satellite Imagery Dataset

8. Project Plan

Activity	Year 2025										Year 2026									
	Aug.		Sept.		Oct.		Nov.		Dec.		Jan.		Feb.		Mar.		Apr.		May	
Literature Review																				
Analysis and Requirements																				
Project Design and Architecture																				
Implementation																				
Testing and Validation																				
Documentation and Write-up																				

9. Signatures

(Khushi Bhasin) (Abhinav Manda) (Aatesh)

(Vanshika Sharma)

(Dr. Arvind Kumar)
Assistant Professor (SG)

Date of Submission: 20th August 2025